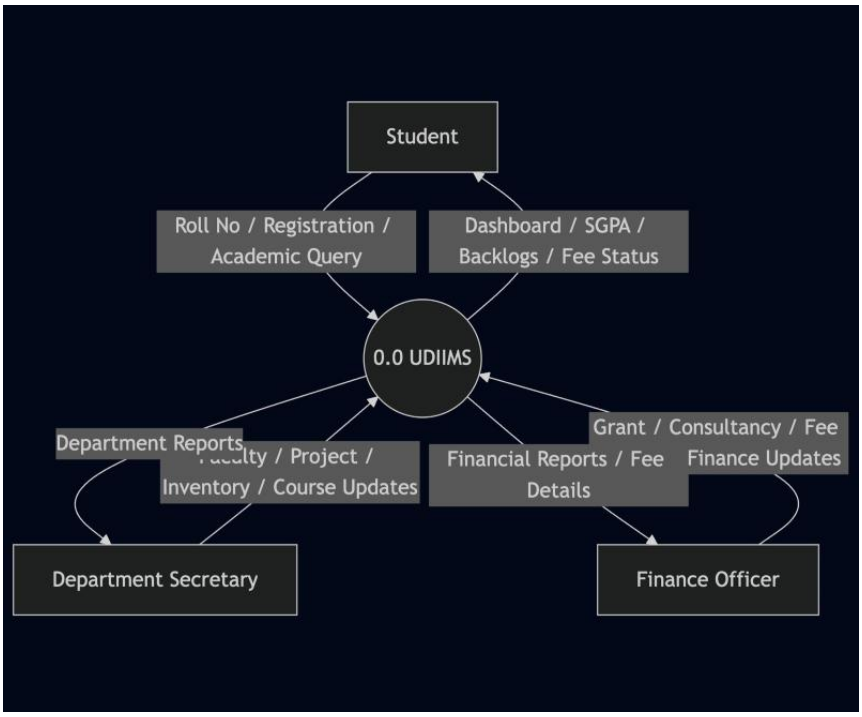
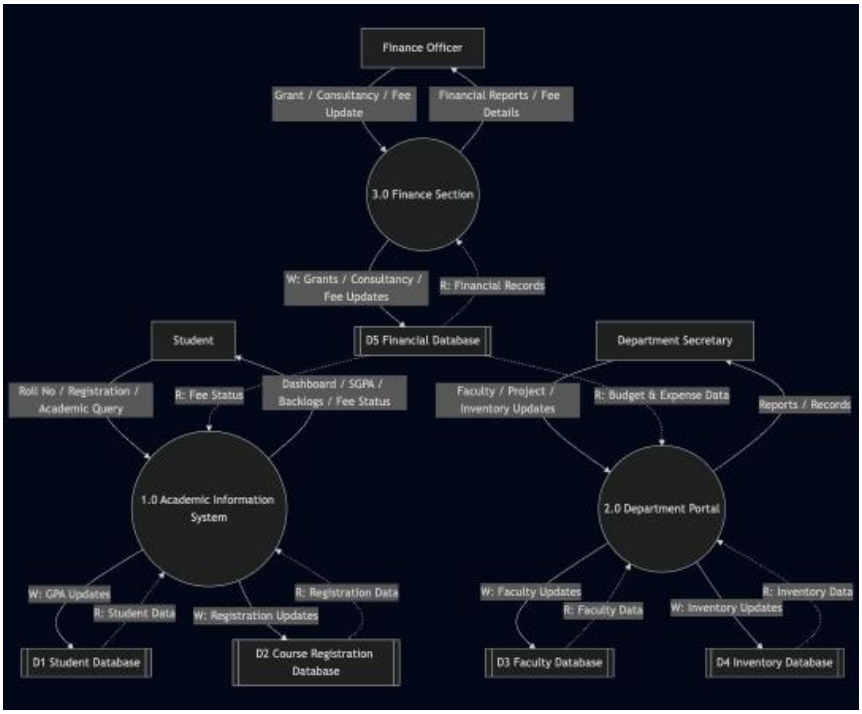
University Department Information & Integration Management System (UDIIMS)

Data Flow Diagram:

Level 0:

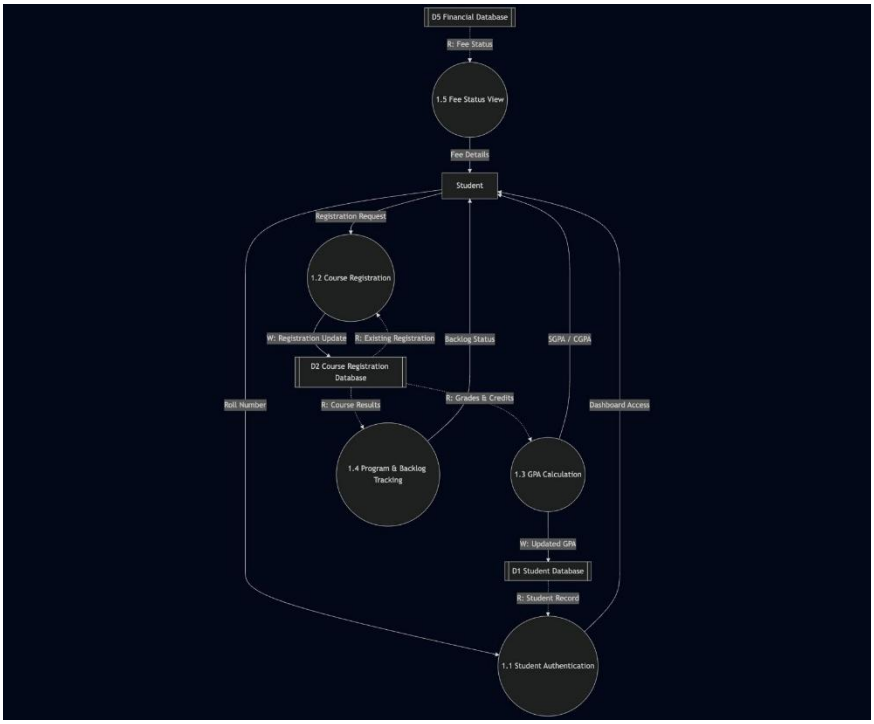


Level 1:

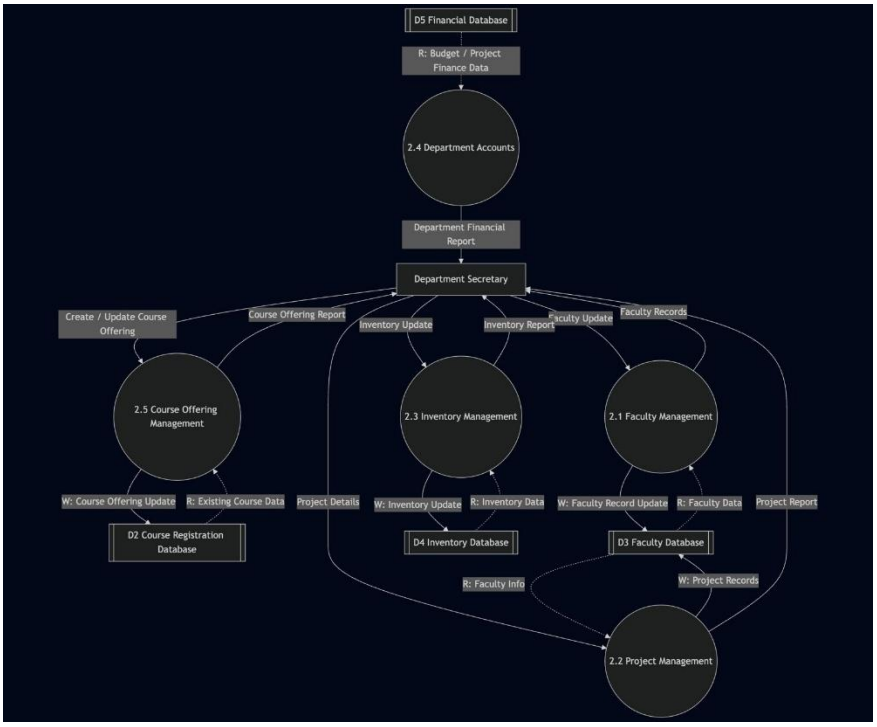


Level 2:

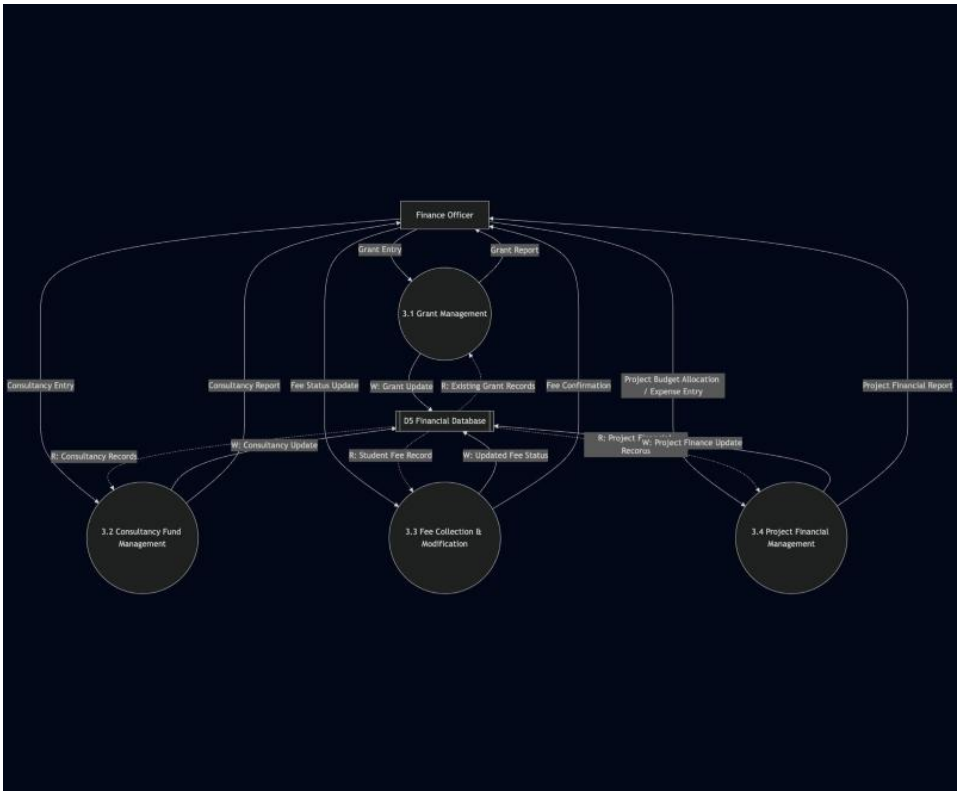
Student Dept:



Acad Dept:



Finance Dept:



Data Dictionary:

1. USER & AUTHENTICATION DATA

login-credentials = student-id + password-authentication
student-id : string
pass-authentication: string

dashboard-access : boolean

student-record = student-id + student-name + program + semester + sgpa + cgpa + backlog-count + dashboard-access
student-id : string
student-name : string
program : string
semester : integer
sgpa : float
cgpa : float
backlog-count : integer

faculty-record = faculty-id + faculty-name + designation + department-name + specialization + active-status
faculty-id : string
faculty-name : string
designation : string
department-name : string
specialization : string
active-status : boolean

users = [student, department-secretary, finance-officer]

2. ACADEMIC INFORMATION DATA

registration-request = student-id + semester-term + {course-code}*
course-code : string
semester-term : string

course-record = course-code + course-name + credit-hours
course-name : string
credit-hours : integer

registration-record = registration-id + student-id + course-record + registration-status + backlog-flag
registration-id : string
registration-status = [active, dropped, completed]
backlog-flag : boolean

gpa-record = student-id + sgpa + cgpa + {gpa-updated-timestamp}*
gpa-updated-timestamp : datetime

backlog-status = student-id + {backlog-course}*

backlog-course = course-code + grade
grade = [A, A-, B, B-, C, C-, D, E, F, F*, I, R, W, Z, AP, AF, AU, S, U, P, NP, TR]

student-dashboard = student-id + gpa-record + backlog-count + fee-status

fee-status = [paid, pending, partial, waived]

3. DEPARTMENT MANAGEMENT DATA

faculty-update = faculty-record

project-details = project-id + project-title + faculty-id + project-budget + project-status
project-id : string
project-title : string
project-budget : float
project-status : string

inventory-record = item-id + item-name + category + quantity + location + condition + assigned-department
item-id : string
item-name : string
category : string
quantity : integer
location : string
condition = [new, good, fair, poor, disposed]
assigned-department : string

course-offering-update = course-code + course-name + credit-hours + semester-term

department-financial-report = department-id + budget-allocation + total-expenses + remaining-balance
department-id : string
budget-allocation : float
total-expenses : float
remaining-balance : float

4. FINANCE SECTION DATA

financial-record = record-id + record-type + amount + transaction-date
record-id : string
record-type = [grant, consultancy, student-fee, project-budget, expense]
amount : float
transaction-date : datetime

grant-entry = grant-id + amount + transaction-date + department-id
grant-id : string

consultancy-entry = consultancy-id + amount + transaction-date + department-id
consultancy-id : string

fee-record = student-id + amount + fee-status + fee-updated-timestamp
fee-updated-timestamp : datetime

project-financial-entry = project-id + project-budget + expense-amount + transaction-date
expense-amount : float

financial-report = {financial-record}*

5. REPORTING DATA

academic-report = {registration-record}*

inventory-report = {inventory-record}*

faculty-report = {faculty-record}*

project-report = {project-details}*

fee-confirmation = student-id + fee-status

6. DATA STORE DEFINITIONS

D1-student-database = {student-record}*

D2-course-registration-database = {registration-record}*

D3-faculty-database = {faculty-record}*

D4-inventory-database = {inventory-record}*

D5-financial-database = {financial-record}*

7. EXTERNAL ENTITY DATA INTERFACE

student-input = [login-credentials, registration-request, student personal details]

student-output = [student-dashboard, backlog-status, fee-confirmation]

department-secretary-input = [faculty-update, project-details, inventory-record, course-offering-update]

department-secretary-output = [faculty-report, project-report, inventory-report, department-financial-report]

finance-officer-input = [grant-entry, consultancy-entry, fee-record, project-financial-entry]

finance-officer-output = [financial-report, fee-confirmation]

8. SYSTEM SUMMARY STRUCTURE

academic-information-system = authentication + course-registration + gpa-calculation + backlog-tracking + fee-status-view

department-portal = authentication+ faculty-management + project-management + inventory-management + course-offering-management + department-accounts

finance-section = authentication+ grant-management + consultancy-management + fee-collection + project-financial-management

external-entities = student + department-secretary + finance-officer